

SOC LOG ANALYSIS REPORT

Host: metasploitable | Auth Log Review

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Report Date: June 09, 2026 | Log Period: Jun 3 – Jun 9, 2026

CLASSIFICATION: INTERNAL / EDUCATIONAL USE ONLY

Report Metadata

Report ID	SOC-RPT-2026-001
Analyst	B. Giri Anoop
Role	SOC Analyst — Student Project (CyArt Internship)
Target Host	metasploitable (192.168.56.20)
Log Source	/var/log/auth.log
Analysis Tool	Custom Python Log Parser
Log Period	Jun 03, 2026 – Jun 09, 2026
Report Generated	Jun 09, 2026
Status	FINAL

1. EXECUTIVE SUMMARY

This report presents the findings of a structured analysis of the Linux authentication log (`/var/log/auth.log`) collected from the Metasploitable 2 host (**192.168.56.20**) covering the period **June 3 – June 9, 2026**. Log data was parsed using a custom Python parser and reviewed for security-relevant events including failed authentication attempts, privilege escalation via sudo, cron-initiated sessions, and anomalous activity patterns.

Key findings are summarised below:

Finding	Severity	Count / Detail
Multiple failed login bursts (brute-force pattern)	HIGH	4 distinct burst events
UNKNOWN user login attempts	HIGH	Multiple across bursts
Sudo poweroff issued repeatedly	MEDIUM	7 instances across sessions
Auth log copied to home directory	MEDIUM	2 sudo cp commands observed
Auth log reviewed via sudo tail	LOW / INFO	3 instances (legitimate admin)
SSH negotiation failure (legacy keys)	INFO	1 — ssh-rsa/ssh-dss only
Routine cron sessions (root)	INFO	Normal scheduled activity

2. SCOPE & METHODOLOGY

Parameter	Value
Log file analysed	<code>/var/log/auth.log</code>
Host	metasploitable (192.168.56.20)
Analysis period	Jun 03 2026 05:26 → Jun 09 2026 05:39
Parser	Custom Python regex-based log parser
Event categories	Login success, Failed login, Sudo activity, Cron activity
MITRE ATT&CK; ver.	v14
Framework	NIST SP 800-61 Rev.2 (Incident Response)

The raw log was parsed into four tagged categories: **[INFO] Login success**, **[ALERT] Failed login**, **[INFO] Sudo Activity**, and **[INFO] Cron Activity**. Events were then correlated temporally to identify patterns indicative of brute-force attempts, privilege abuse, or evidence tampering.

3. EVENT STATISTICS

Event Type	Count	% of Total
Login success (all)	56	71.8%
— Interactive (msfadmin)	16	20.5%
— CRON (root)	38	48.7%

— Sudo session open	18	23.1%
Failed login [ALERT]	14	17.9%
Sudo activity (commands)	9	11.5%
Cron closed sessions	14	—
TOTAL parsed events	78	100%

4. DETAILED FINDINGS

4.1 Failed Login Bursts — Suspected Brute-Force Activity

Severity: **HIGH** **MITRE ATT&CK:** T1110.001 – Brute Force: Password Guessing

Four distinct burst events were identified where consecutive failed logins occurred within seconds, followed by a successful authentication. This pattern is consistent with manual password guessing or a low-rate brute-force attempt. Notably, several failures target **UNKNOWN** usernames, suggesting enumeration alongside guessing.

Burst #	Time Window	Failures	Users Targeted	Success After?
1	Jun 8 01:10:22	1	msfadmin	Yes — 01:10:30
2	Jun 8 01:32:16–01:32:43	4	msfadmin, UNKNOWN (x2)	Yes — 01:33:12
3	Jun 8 01:41:01–01:41:39	4	msfadmin, UNKNOWN (x2)	Yes — 01:42:07
4	Jun 8 08:53:18–08:53:43	3	msfadmin (x2), UNKNOWN	Yes — 08:53:49
5	Jun 9 05:19:56–05:20:10	3	UNKNOWN, msfadmin (x2)	Yes — 05:20:15

Notable: Burst #2 and #3 occurred within 9 minutes of each other on Jun 8 (01:32 and 01:41), indicating persistent access attempts in a short window. All bursts terminate with successful login — possible valid credentials were eventually supplied, or the attacker already knew the password and mis-typed.

4.2 Sudo Privilege Escalation — Command Analysis

Severity: **MEDIUM** **MITRE ATT&CK:** T1548.003 – Abuse Elevation Control Mechanism: Sudo and Sudo Caching

All sudo commands were executed by **msfadmin** from **TTY=tty1** (physical/virtual console), escalating to root. Nine distinct sudo command executions were identified across the log period.

Timestamp	Command Executed	Category	Risk
Jun 3 05:26	(session open – no command recorded)	Session	INFO
Jun 8 00:18	/sbin/poweroff	Shutdown	MEDIUM
Jun 8 00:53	/usr/bin/tail -30 /var/log/auth.log	Log Review	LOW
Jun 8 01:09	/usr/bin/tail -50 /var/log/auth.log	Log Review	LOW

Jun 8 01:18	/bin/cp /var/log/auth.log /home/msfadmin/	Log Copy	MEDIUM
Jun 8 01:42	/bin/cp var/log/auth.log /home/msfadmin/	Log Copy (err)	MEDIUM
Jun 8 01:43	/bin/cp /var/log/auth.log /home/msfadmin/	Log Copy	MEDIUM
Jun 8 02:23	/sbin/poweroff	Shutdown	MEDIUM
Jun 8 09:33	/sbin/poweroff	Shutdown	MEDIUM
Jun 8 11:47	/sbin/poweroff	Shutdown	MEDIUM
Jun 9 02:58	/sbin/poweroff	Shutdown	MEDIUM
Jun 9 03:21	/sbin/poweroff	Shutdown	MEDIUM
Jun 9 03:23	/sbin/poweroff	Shutdown	MEDIUM
Jun 9 03:32	/sbin/poweroff	Shutdown	MEDIUM
Jun 9 04:41	/sbin/poweroff	Shutdown	MEDIUM

Key Observation — Log Copying: The auth log was copied three times to **/home/msfadmin/auth.log**. One attempt used a relative path (**var/log/auth.log** — missing leading slash) suggesting either a typo or an inexperienced operator. Copying logs to user home directory may indicate evidence collection for exfiltration OR a student/admin reviewing logs offline. In a real environment this would warrant escalation.

Key Observation — Repeated Shutdown: **sudo /sbin/poweroff** was executed 8 times across Jun 8–9. While consistent with a lab environment being powered down after sessions, repeated escalated shutdowns should be flagged in production environments.

4.3 SSH Connection Failure — Legacy Key Negotiation

Severity: INFO **MITRE ATT&CK:** T1021.004 – Remote Services: SSH

A connection attempt was made from the analyst's Kali host (**karna@Bluejack**) to **192.168.56.20:22**. The attempt failed at key negotiation because the Metasploitable 2 SSH server only advertises legacy host key types (**ssh-rsa**, **ssh-dss**), which modern OpenSSH clients reject by default.

Error observed:

```
Unable to negotiate with 192.168.56.20 port 22: no matching host key type found. Their offer:
ssh-rsa, ssh-dss
```

Remediation (Lab): Add **HostKeyAlgorithms +ssh-rsa** and **PubkeyAcceptedAlgorithms +ssh-rsa** to **~/.ssh/config** on the Kali host to allow connection for testing purposes only.

4.4 Cron Sessions — Root Scheduled Tasks

Severity: INFO / NORMAL

Numerous cron-initiated sessions for **root** were observed throughout the log period. All sessions open and close in the same second (or within one second), consistent with short-running scheduled tasks (e.g., logrotate, cleanup scripts). Cron patterns at **:09, :17, :39** of each hour are consistent with default Metasploitable 2 cron configuration. No anomalous cron entries detected.

5. INCIDENT TIMELINE (Jun 8 Focus)

June 8 represents the highest-activity day with multiple failed login bursts, log access, log copying, and repeated shutdown commands. Timeline below:

Time (Jun 8)	Event	Severity
00:08	msfadmin interactive login	INFO
00:18	sudo poweroff issued	MEDIUM
00:19	msfadmin login again (post-reboot)	INFO
00:53	sudo tail -30 /var/log/auth.log	LOW
01:09	sudo tail -50 /var/log/auth.log	LOW
01:10	FAILED LOGIN x1 then success	HIGH
01:11	sudo tail auth.log again post-failure	LOW
01:18	sudo cp auth.log → home dir (log copy #1)	MEDIUM
01:32–01:33	FAILED LOGIN BURST x4 → success (Burst #2)	HIGH
01:41–01:42	FAILED LOGIN BURST x4 → success (Burst #3)	HIGH
01:42	sudo cp auth.log (wrong path)	MEDIUM
01:43	sudo cp auth.log (corrected path)	MEDIUM
02:23	sudo poweroff	MEDIUM
08:51	msfadmin login	INFO
08:53	FAILED LOGIN x3 → success (Burst #4)	HIGH
09:33	sudo poweroff	MEDIUM
11:47	sudo poweroff — end of Jun 8 session	MEDIUM

6. MITRE ATT&CK; MAPPING

Technique ID	Technique Name	Tactic	Observed Evidence
T1110.001	Brute Force: Password Guessing	Credential Access	4–5 burst events on pts/1
T1078	Valid Accounts	Defense Evasion	msfadmin used post-failure
T1548.003	Sudo and Sudo Caching	Privilege Escalation	Multiple sudo escalations
T1005	Data from Local System	Collection	Auth log cp to home dir
T1070.002	Indicator Removal: Clear Linux Logs	Defense Evasion	Log copy — possible staging
T1021.004	Remote Services: SSH	Lateral Movement	SSH attempt (failed — legacy)
T1059.004	Unix Shell	Execution	Shell commands via sudo tty1

7. INDICATORS OF COMPROMISE (IOCs)

IOC Type	Value / Pattern	Context
Host	metasploitable (192.168.56.20)	Target system
Username	msfadmin	Primary account targeted & used
Username	UNKNOWN	Non-existent user — enumeration attempt
TTY	pts/1	All failed/suspicious logins on this TTY
Path	/home/msfadmin/auth.log	Auth log staging location
Command	/bin/cp /var/log/auth.log /home/msfadmin/	Log exfiltration command
Command	/sbin/poweroff	Repeated privilege shutdown
Port	22 (SSH)	Legacy key negotiation failure
Log Pattern	FAILED LOGIN (N) on 'pts/1' FOR `UNKNOWN`	User enumeration signature

8. RISK ASSESSMENT

Finding	Likelihood	Impact	Risk Level
Brute-force on local TTY	HIGH	HIGH	CRITICAL
Auth log copied to user home	CONFIRMED	MEDIUM	HIGH
Repeated sudo shutdowns	CONFIRMED	MEDIUM	MEDIUM
UNKNOWN user enumeration	HIGH	MEDIUM	HIGH
SSH legacy key (no auth reached)	LOW	LOW	LOW
Cron root sessions	NORMAL	LOW	INFO

9. RECOMMENDATIONS

R1	HIGH	Enforce account lockout policy
Implement pam_tally2 or pam_faillock to lock accounts after 3–5 consecutive failures. Prevents brute-force success even with weak passwords.		
R2	HIGH	Monitor pts/1 for failed login patterns
Create a Wazuh/SIEM rule to alert on ≥ 3 failed logins within 60 seconds on any single TTY. Map to rule 5710 (multiple auth failures).		
R3	MEDIUM	Audit sudo cp of sensitive log files
Any sudo copy of /var/log/* to user-owned directories should trigger an alert (potential log staging for exfiltration). Use auditd rule: -a always,exit -F arch=b64 -S execve -F exe=/bin/cp.		
R4	MEDIUM	Restrict sudo poweroff to specific maintenance windows

Repeated escalated shutdowns outside scheduled windows indicate either an attacker or sloppy lab hygiene. Apply sudoers time restriction.

R5	LOW	Upgrade SSH host keys on Metasploitable (lab note)
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SSH server only offers ssh-rsa/ssh-dss. While expected in this intentionally vulnerable lab, document this as a known weak configuration.

R6	INFO	Centralise log shipping
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Ship auth.log to a SIEM in real-time so local log copy/deletion does not destroy forensic evidence. Use Filebeat or rsyslog forwarding.

10. CONCLUSION

The authentication log analysis of the Metasploitable 2 host reveals a pattern consistent with **manual or low-rate brute-force activity on the local console (pts/1)**, combined with privilege escalation, log access, and log staging. All suspicious activity traces back to the **msfadmin** account, which is the default credential on this intentionally vulnerable platform.

In a production environment, the combination of repeated authentication failures followed by success, paired with immediate sudo log access and log copying, would constitute a **Tier 2 security incident** warranting immediate investigation, account suspension, and forensic imaging of the affected host.

This analysis was conducted as part of the **CyArt SOC Analyst Internship** student project. All findings are based on the provided auth.log extract and the custom parser output reviewed during this engagement.

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